# Linear Circuits 2

# Thevenin’s Theorem Lab Report

Group Members:

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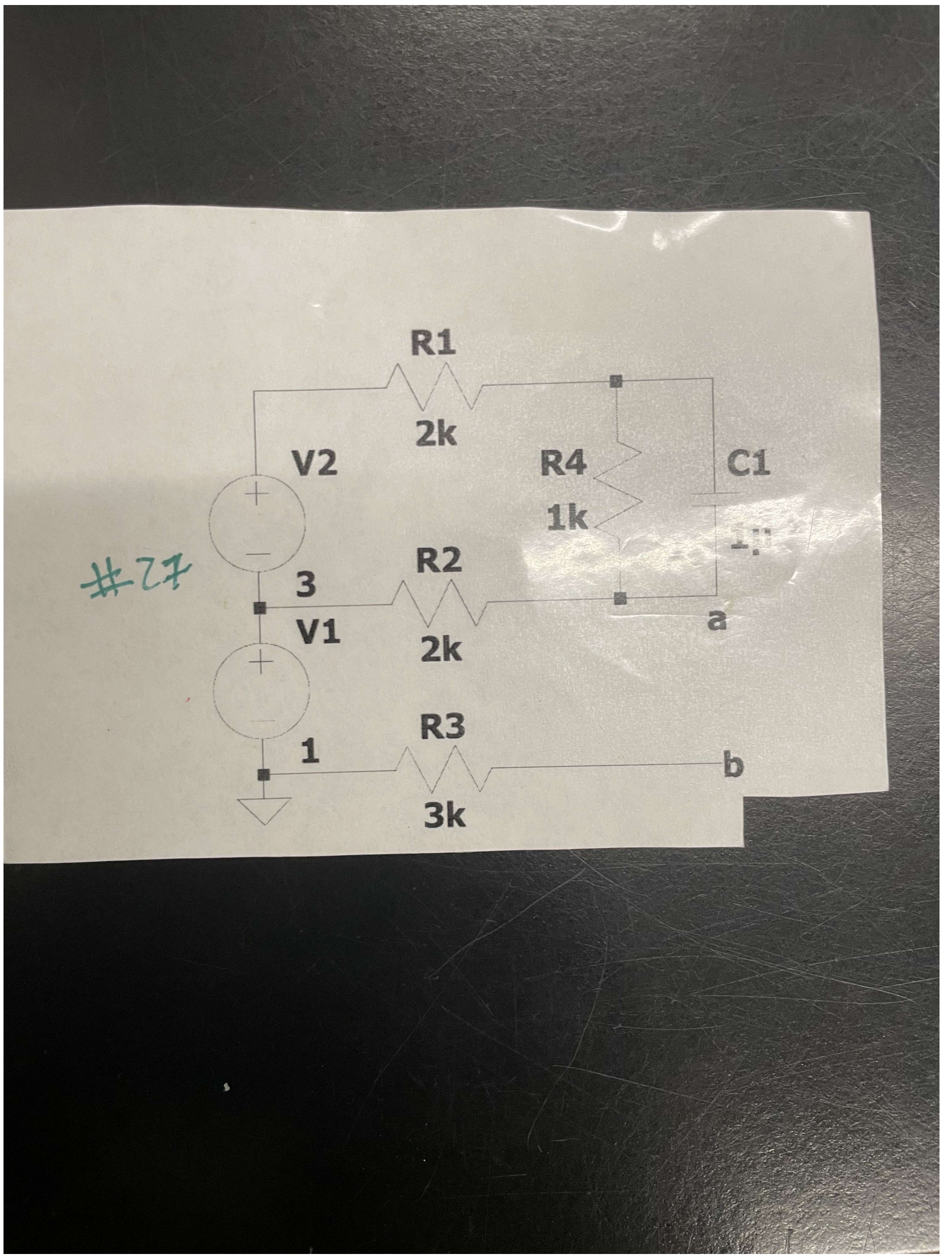
## Objectives:

The objective of the project was to apply our knowledge of Thevenin and Norton equivalent circuit that was learnt in Linear Circuits 1, and verify those results in real world setting, via both simulation of the circuit in LTSpice as well as physically proving the circuits correct on a breadboard with the respective components.

The circuit that was to be verified is shown below:



# Equipment:

- Digital Multimeter (DMM)
- Rohde & Schwarz Power Supply Series
- 1u Capacitor
- One 1k Resistor
- Two 2k Resistors
- One 3k Resistor

# Hand Calculations:

To implement this circuit, we first use LTSPice to create it exactly as was shown via the picture above. Once it was created, it should appear as the following:

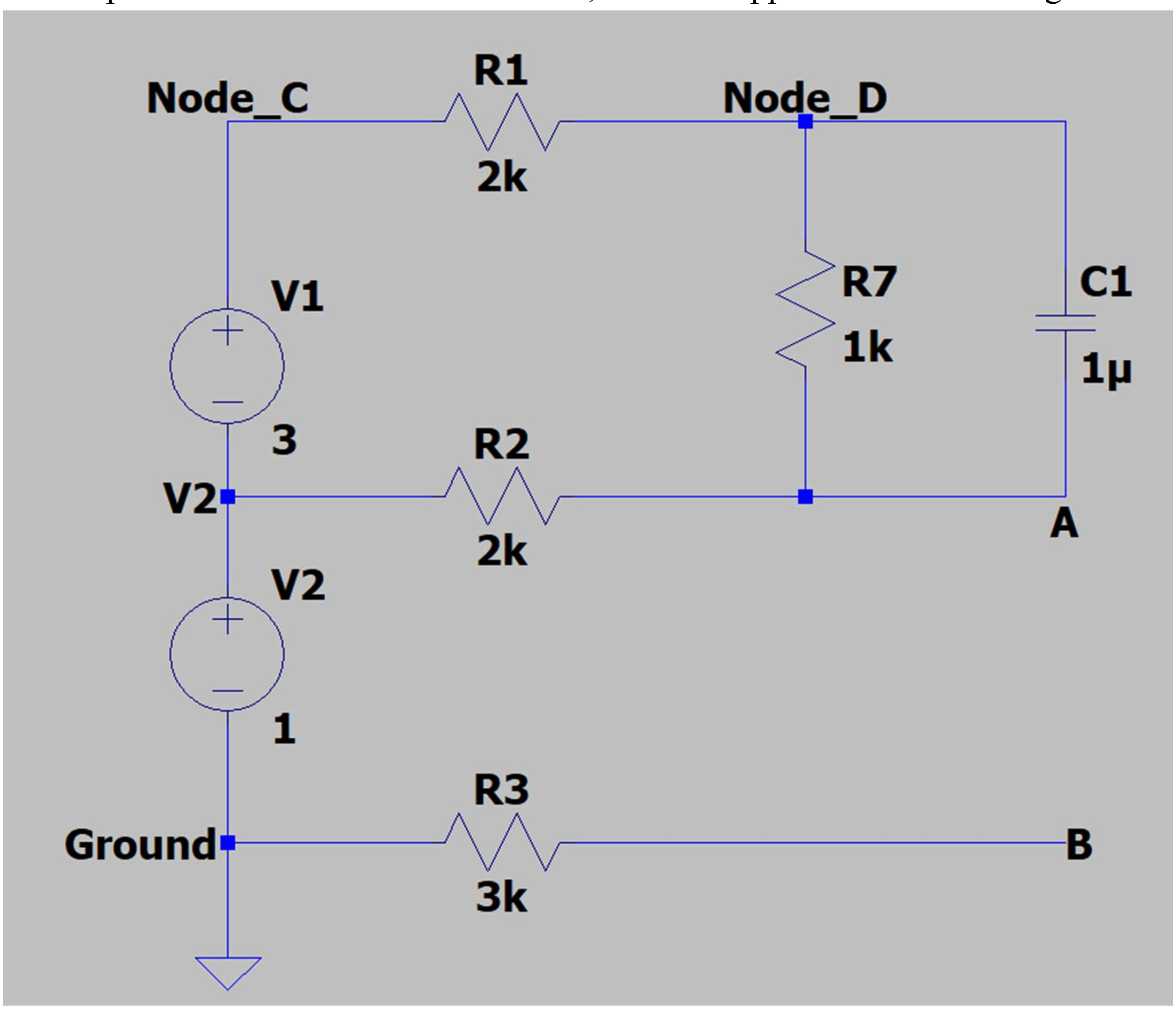



Figure 1: Original Circuit

**When the circuit is shorted across Node a and b:**

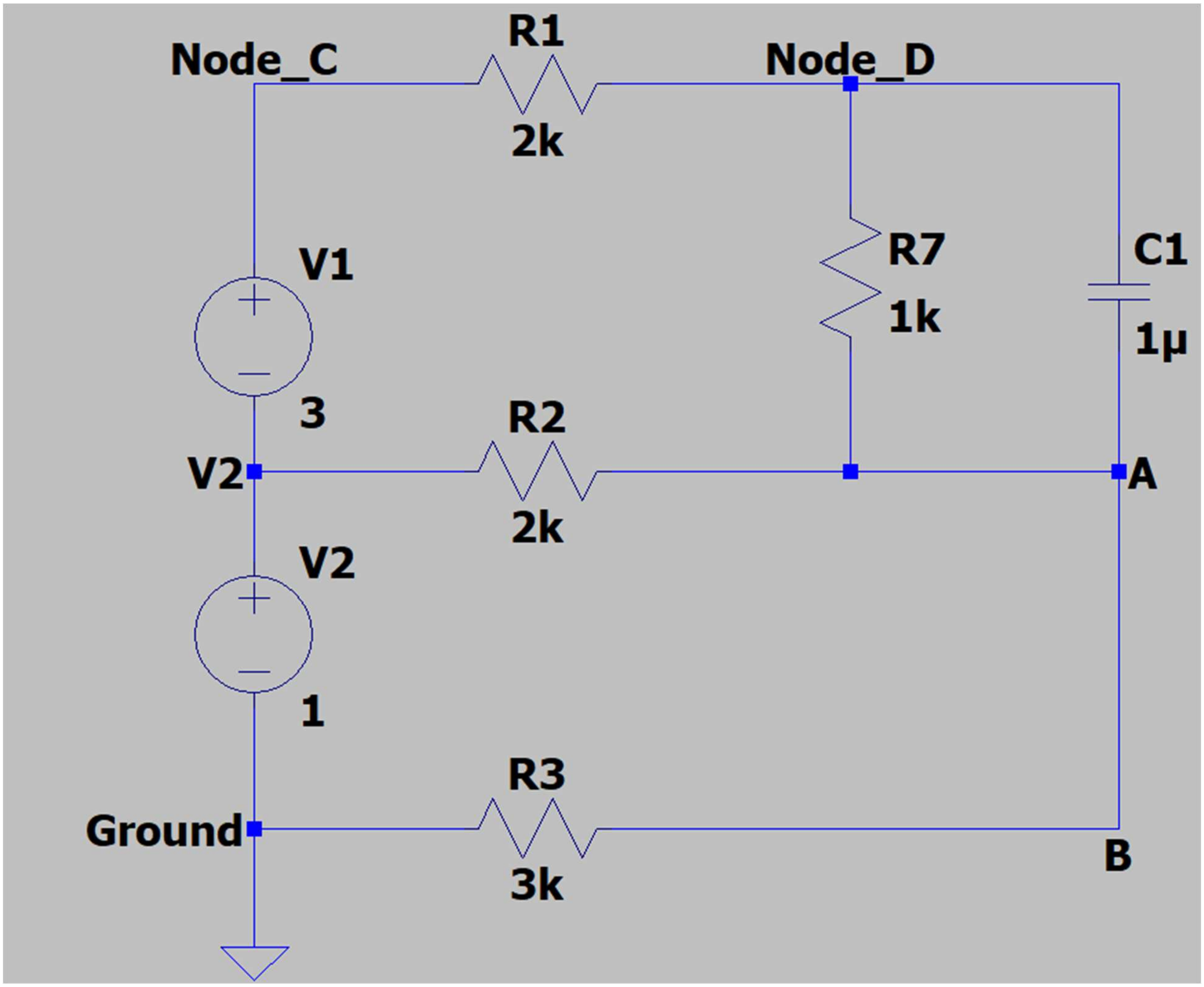



Figure 2: Shorted Circuit

In the simulation, we found the Thevenin Voltage which would be the difference in voltage between Node A & B with the open ciruit. As well as the Norton Current by measuring the current running through the shorted wire through Nodes A & B. After this, we then performed a hand analysis to confirm values, of which the steps are shown below:

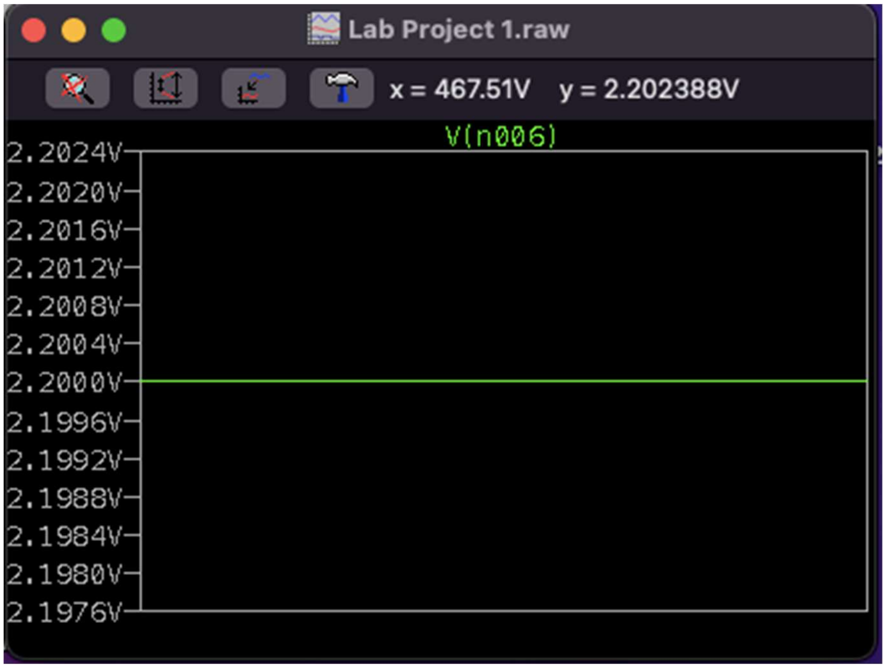



Figure 3: Voltage Thevenin from Simulation

1) First, remember Ohm's law we know V=IR, and R=V/I. So, to get an idea of each value, we first use the original circuit simulation to measure Vth. Resulting in 2.2V(shown above).
2) After this, the next value we can obtain is Rth, which we can do multiple ways. The way we chose to do this is by shorting all the independent sources and replacing the open circuit between Node A & B with a test voltage of 1V.
3) Next, using the simulation or KCL, we measure the current going across that voltage source which is equivalent to 238.09uA.
4) Once this is done, simply use Ohms law to get Rth by calculating 1V/238.09uA. The result being 4200.09 Ohms.
5) This gives us all the components we need to create the Thevenin Equivalent Model, but to convert to Norton it's fairly simple.
6) To get the current needed, divide Vth by Rth which yeilds the result 5.24*10^-4A and put a 4200 Ohm resistor in parallel.
7) You also get the same result by shorting the wire between A & B and just measuring/solving the current.

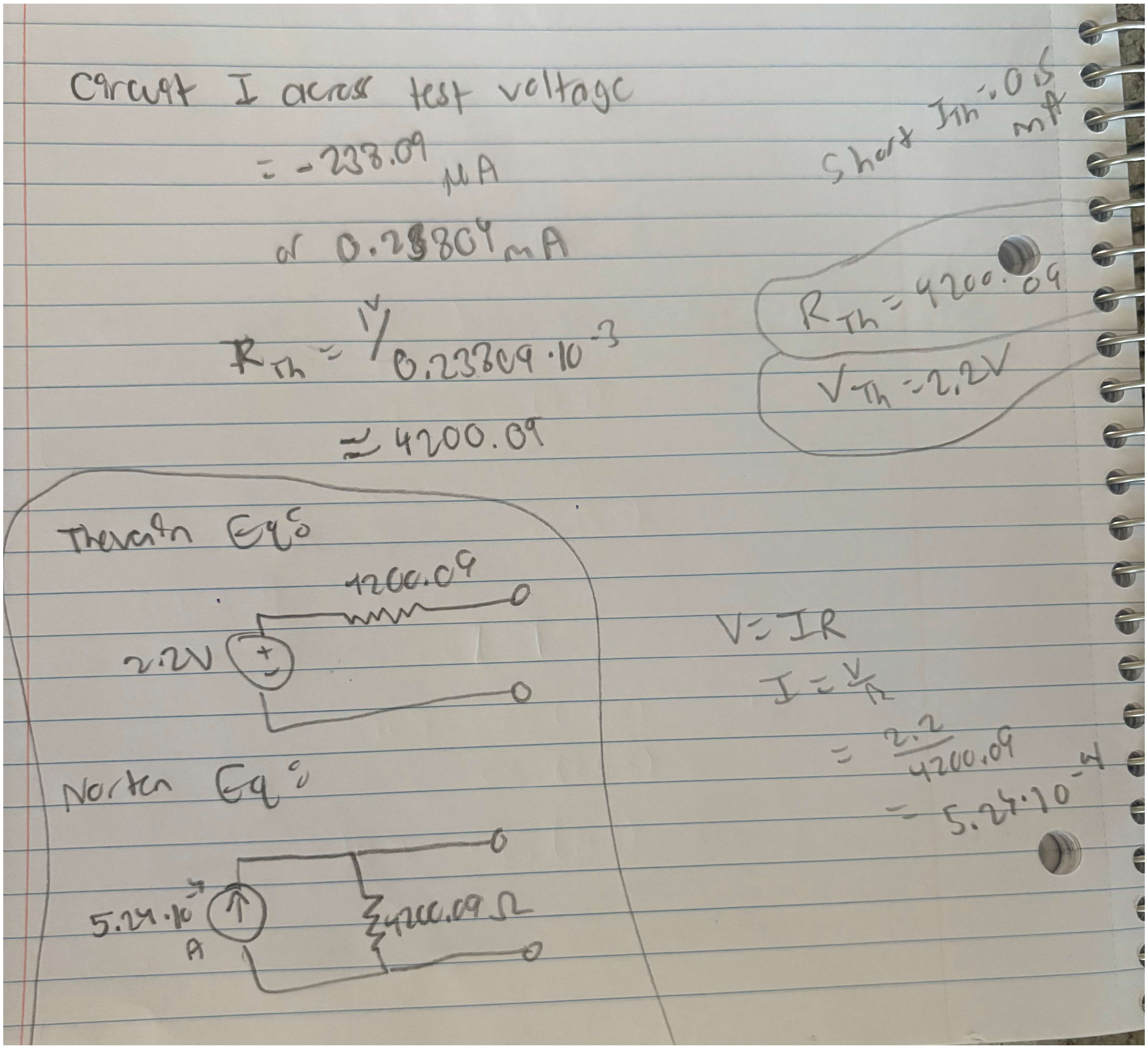



Figure 4: Hand Calculations

These results are also shown in the Simulation Results section as well.

After this, we then put the circuit onto the breadboard as shown below and verify it with a Digital Multimeter.

## Simulation:

For the simulation, we decided to build multiple different circuits to assist us in finding each of our values.

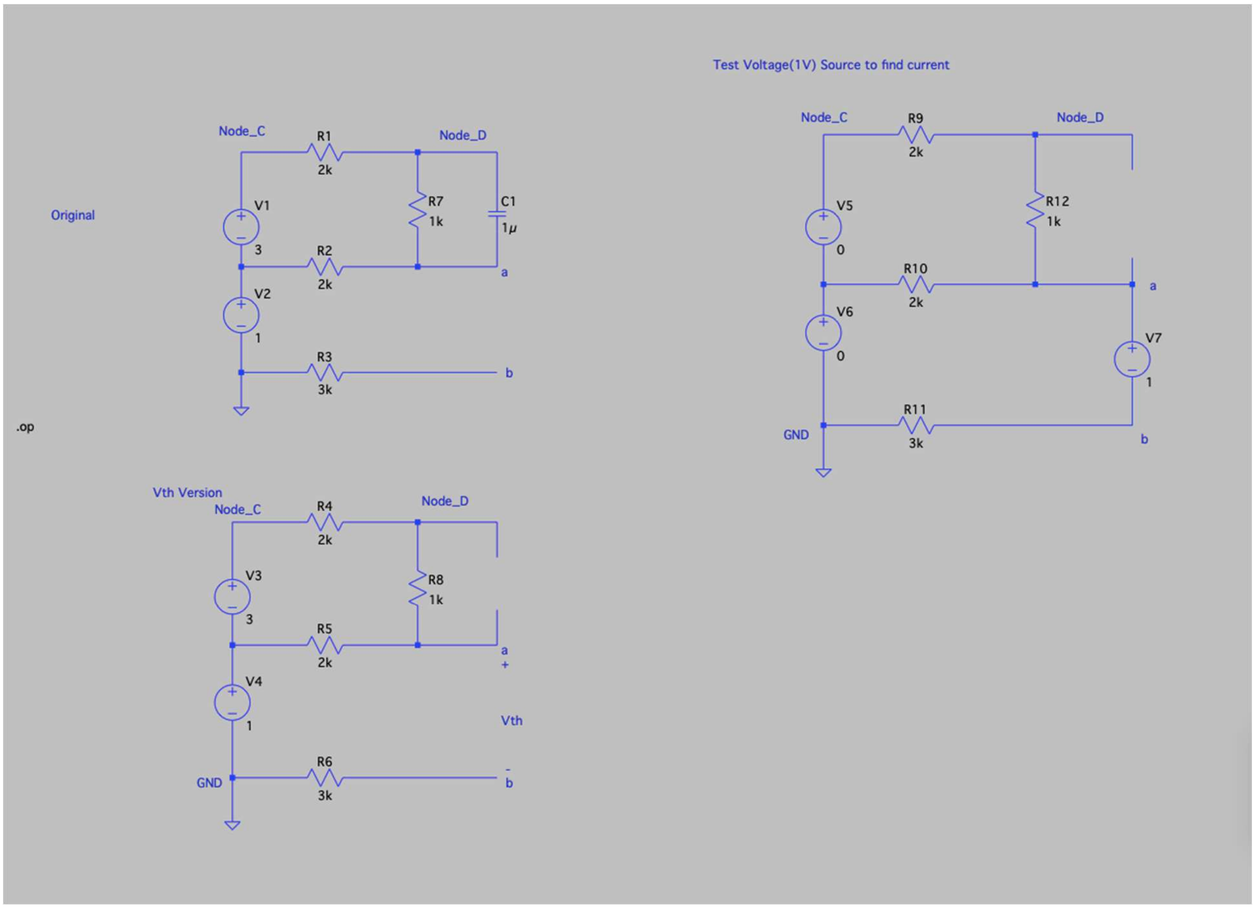



Figure 5: Simulated Circuits

Continuing to our simulation results, with the following image shown below for the original circuit:

C:\Users\monde\Downloads\Lab Project 1.net

```
      --- Operating Point ---

V(a):            2.2            voltage
V(b):            0              voltage
V(node_c):       4              voltage
V(node_d):       2.8            voltage
V(v2):           1              voltage
I(C1):           6e-19          device_current
I(R1):           -0.0006        device_current
I(R2):           0.0006         device_current
I(R3):           0              device_current
I(R7):           0.0006         device_current
I(V1):           -0.0006        device_current
I(V2):           4.33681e-19    device_current
```

Figure 6: Original Circuit Simulation Results

**Shown above are the results for our first circuit. The important value to gather from this is the voltage V(a), which is 2.2V. (Vth)**

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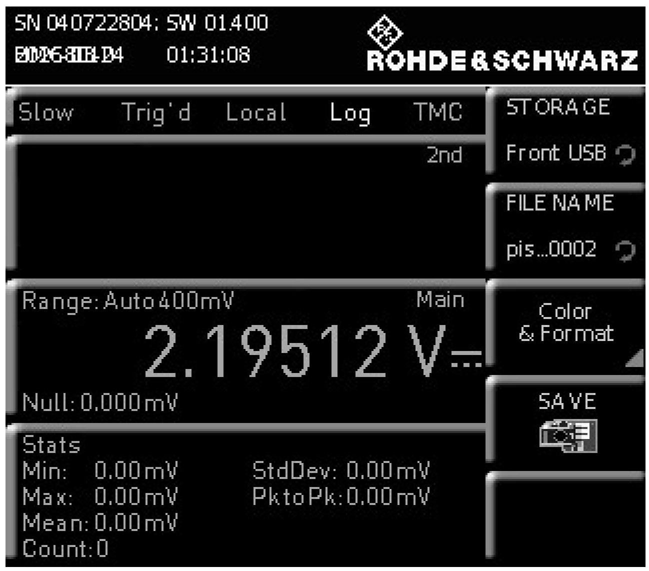
```
      --- Operating Point ---

V(b):            1.57143        voltage
V(node_c):       4              voltage
V(node_d):       2.38095        voltage
V(v2):           1              voltage
I(C1):           8.09524e-19    device_current
I(R1):           -0.000809524   device_current
I(R2):           0.000285714    device_current
I(R3):           0.00052381     device_current
I(R7):           0.000809524    device_current
I(V1):           -0.000809524   device_current
I(V2):           -0.00052381    device_current
```

Figure 7: Shorted Circuit Simulation Results

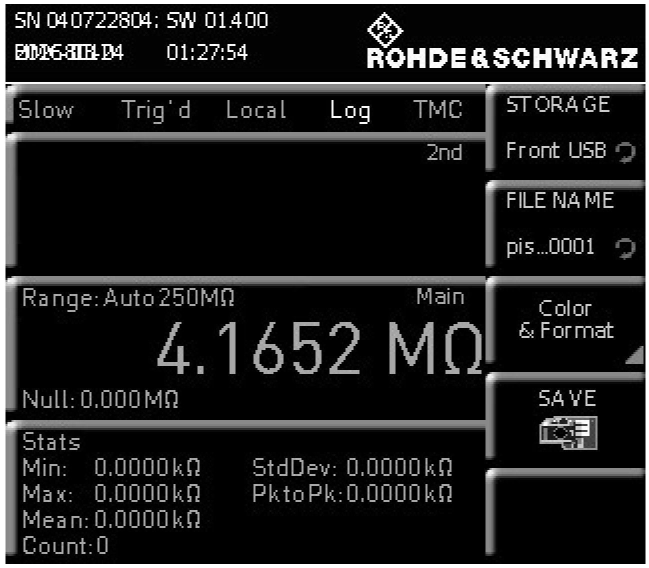
Here in this image, are the values obtained from the short circuit version. If you wanted to find Isc, it's the equivalent of the current across R3, approximately 5*10^-4A. Plugging both the Vth and Isc into Ohm's law will give you Rth.

## Experiment:

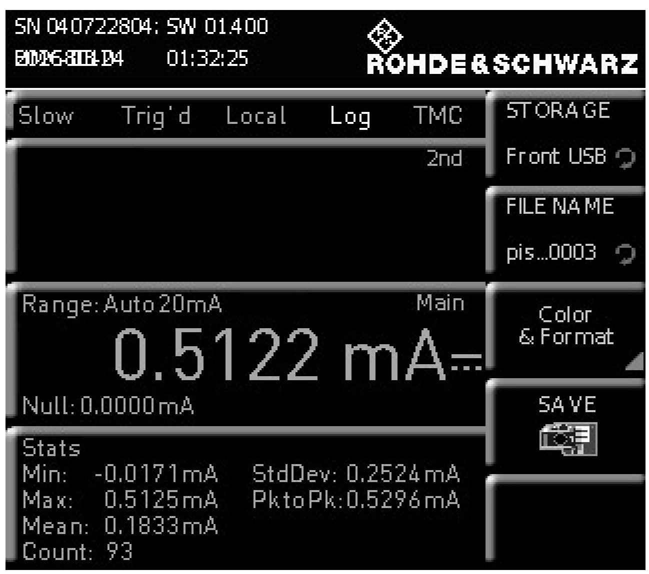
For the experimental results, with the breadboard shown above we measure the voltage between nodes A and B to get the following voltage:




For the equivalent resistance, we turn off all the independent sources and use the probes of the Digital Multimeter to get the following equivalent resistance between nodes A and B:




For the current, we broke the circuit to measure the current traveling across the nodes A and B to get the following current:




# Results:

Both the experimental and simulation results are almost identical to each other, with the only discrepancy being our Thevenin resistance value. After further inspection our DMM picture has the result 4.16M ohms rather than 4.16K ohms.

This could possibly be the result of a broken circuit accounting for higher resistance than normal given that our calculations and simulations both prove that 4.16K ohms is the correct value. Overall, the results do align with each other, and a simple mistake is most likely what caused the difference in resistance values given the digits are the same, just on a larger scale.

## Conclusion:

This project has been successful in validating the principles of Thevenin and Norton's theorems via comparing the hands-on calculations, virtual simulations, and real-world experimentation. The initial hand analysis determined that the Thevenin voltage to be 2.2V and the Norton current to be 0.5mA, values whose validation was strengthened by the LTSpice simulation results of 2.2 V and 0.52 mA, respectively. In the experimental phase, the physical circuit yielded a measured voltage of 2.195 V and a Norton current of 0.512 mA. Overall, the close accuracy across all three validations confirms the application of circuit theory, making this project a success.